SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools. *(BL4,BL5)*

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

Code of Conduct:

19/20

I Abarajithan S I 192521299 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.
3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance.Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and

stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized	Handles most cases; reasonable	Handles basic cases only	Incomplete or inefficient

		approach	efficiency		

1. Throughput and Packet loss

Pseudo code:

```
Begin
Data = 8000
Time = 4
Sent = 200
Received = 180
If time = 0 or sent = 0 then
    Print "Invalid input"
Else
    Throughtput = Data / time
    Loss = ((sent - received) / sent) * 100
    Print "throughput =", throughput, "bps"
    Print "Packet loss =", Loss, "%"
    If throughput > 1500 And loss < 10 then
        Print "Network is inefficient"
    End if
End if
End
```

Sample output :

Throughput = 2000 bps

packet loss = 10 %.

Network is inefficient

Calculation :

→ Throughput = 8000 / 4 = 2000 bps

→ packet loss = ((200 - 180) / 200) × 1(

= 10 %.

2. Bandwidth Monitoring

Pseudo code :

```
Begin
While true
    For each link
        usage = Get - bandwidth ( link )
        Print " link usage = ", usage , " % "
        If usage > 80 then
            Print " link is overloaded "
            Print " Switch to backup link "
        Else
            Print " link is normal "
        End if
    End for
    wait 10 seconds
End while
End
```

Sample output :

Link usage = 65%.
Link is Normal

Link usage = 85%.
Link is overloaded
Switch to backup link

Link usage = 72%.
Link is normal

3. Best Network Path

```
Begin
While true
    For each link
        Bandwidth = Get - Bandwidth ( link )
        Delay = Get - Delay ( link )
        Usage = Get - utilization ( link )
        Score = Bandwidth - Delay - usage
        If usage > 80 or link failed then
            Print " link is overloaded or failed"
            Print " select Alternate link"
        End if
    End for
    select link with highest score
```

Print "best path selected"

wait 10 seconds

End while

End

Sample output :

Link 1 :

Bandwidth = 100

Delay = 10

usage = 40

Score = 50

Link 2 :

Bandwidth = 80

Delay = 5

Usage = 30

Score = 45

Link 3 :

Bandwidth = 120

Delay = 20

Usage = 50

Score = 50

Best path selected , Link 1